

Fake News Detection on the LIAR Dataset

Deep Learning I — Final Project. Section 1, Topic 11 (Task 1).

Problem

Fact-checking does not scale. Sites like PolitiFact check claims by hand, and there are far more claims being made than people to check them. The question we looked at is whether a model can do a useful first pass: given a short political statement, flag it as likely fake or likely true so a human reviewer can prioritise.

We used the LIAR dataset (Wang, 2017), which contains 12,836 statements from PolitiFact, each rated on a six-point scale from "pants-fire" to "true". Following common practice we collapse this into a binary label: the three least-truthful ratings (pants-fire, false, barely-true) become FAKE, and the three most-truthful (half-true, mostly-true, true) become REAL. After this mapping the data is mildly imbalanced, about 44% FAKE, and we keep the dataset's official train / validation / test split (10,269 / 1,284 / 1,283).

Two things make this hard, and both shaped how we approached it. First, the statements are short, with a median of 17 words, so there is very little text to reason about. Second, the binary mapping puts genuinely borderline claims on the decision boundary: a "barely-true" and a "half-true" are almost the same thing, but one is labelled FAKE and the other REAL. We report precision and recall on the FAKE class throughout, because for a triage tool what matters is how many flagged claims are actually fake (precision) and how many of the fakes get caught (recall).

Approach

We built everything in PyTorch and trained every neural model with the same loop (Adam, binary cross-entropy loss, early stopping on validation F1). Keeping the training procedure identical means any difference we see comes from the model, not from how it was trained. All vectorizers, scalers and vocabularies are fit on the training set only, and every run is seeded so the results reproduce.

The brief asks for two models, and those are the core of the project.

- **An MLP on hand-crafted text features.** Each statement becomes a TF-IDF vector (unigrams and bigrams, 5,000 terms) plus eleven simple linguistic features such as length, punctuation counts, sentiment and lexical diversity. A small two-layer network reads that.
- **A BiLSTM on the raw text.** We tokenise the statement, learn word embeddings from scratch, and run a bidirectional LSTM over the sequence.

We made a point of keeping this comparison fair. In the head-to-head, the MLP sees only *text* features, the same information the LSTM has. LIAR also ships speaker metadata (party and the speaker's past truthfulness record), but we hold that out of the main comparison and study it on its own, because handing it to one model and not the other would not be a fair test of the two architectures.

We then added two models beyond the brief because they made the findings more complete:

- **A fine-tuned DistilBERT** on the text, to check whether a pretrained model changes the picture.
- **A hybrid** that takes the BiLSTM's text representation, concatenates it with the speaker metadata, and passes the combination to a final classifier. In effect it fuses the two required models.

Experiments

We ran four controlled experiments and then a single comparison of every model.

Experiment 1 — MLP vs. BiLSTM on text. Averaged over five seeds, the two models are tied.

Model (text only)	Precision	Recall	F1	ROC-AUC
MLP (TF-IDF + linguistic)	0.53	0.50	0.514 ± 0.010	0.62
BiLSTM	0.54	0.52	0.530 ± 0.018	0.64

McNemar's test on their predictions gives $p = 0.69$, so the small gap is noise. Both barely clear a logistic-regression baseline (F1 0.49). On text alone, the sequence model's extra machinery does not buy anything here.

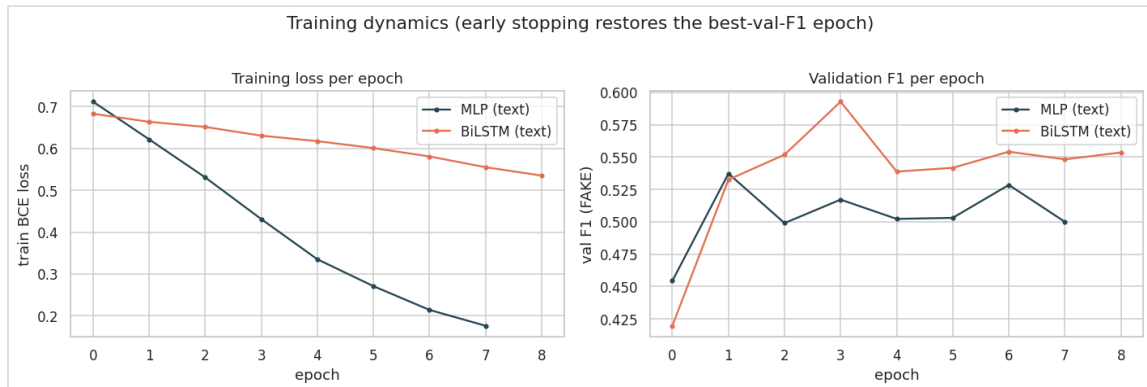


Figure 1. Training loss and validation F1 per epoch; early stopping keeps the best-validation epoch.

Experiment 2 — where does the signal come from? We kept the MLP fixed and changed only its inputs.

Features	F1	ROC-AUC
TF-IDF only	0.509	0.65
+ linguistic	0.520	0.62
+ metadata	0.563	0.67
all	0.567	0.67
metadata only (12)	0.641	0.75

Text features top out around F1 0.51. Adding the speaker metadata lifts that to 0.56, and the result that surprised us most: metadata on its own, just twelve numbers, scores 0.64 with an AUC of 0.75 and beats every text-based model. Most of what looks like fake-news detection on LIAR is really predicting whether this particular speaker tends to lie.

Experiment 3 — LSTM design choices. A unidirectional LSTM slightly beat the bidirectional one (0.557 vs. 0.520), hidden size barely mattered (0.52–0.53 across 64, 128 and 256 units), and initialising the embeddings with pretrained GloVe vectors made no real difference (0.540 vs. 0.544 for random). All three point the same way: the model is not being held back by its capacity or its word vectors.

Experiment 4 — threshold and class balance. Moving the decision threshold trades precision for recall, as expected. We can push FAKE recall from 0.50 up to 0.95, but precision drops to 0.45 and the AUC does not move. The operating point is a choice you make for the application, not a genuine improvement in the model.

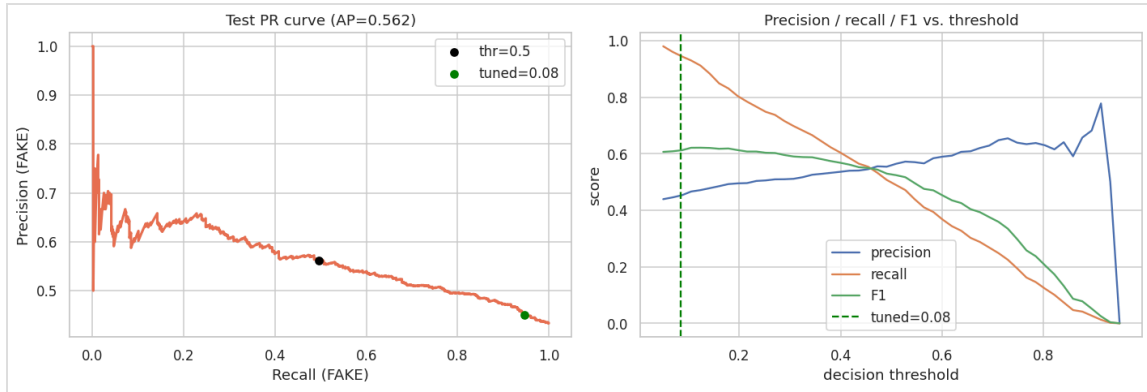


Figure 2. Test precision-recall curve, and precision / recall / F1 as the decision threshold varies.

Final comparison. Every model on the same test set, with bootstrap 95% confidence intervals on F1.

Model	Precision	Recall	F1	ROC-AUC	Accuracy	Params
LogReg (text)	0.57	0.43	0.49	0.65	0.61	5K
MLP (text)	0.56	0.47	0.51	0.65	0.61	1.3M
MLP (text + meta)	0.57	0.51	0.54	0.66	0.62	1.3M
BiLSTM (text)	0.53	0.56	0.55	0.64	0.60	1.2M
Hybrid (text + meta)	0.56	0.59	0.58	0.67	0.62	1.2M
DistilBERT (text)	0.63	0.57	0.60	0.69	0.67	66M

DistilBERT comes out on top, but it costs 66 million parameters and about three minutes of training, against roughly a million parameters and fifteen seconds for the others.

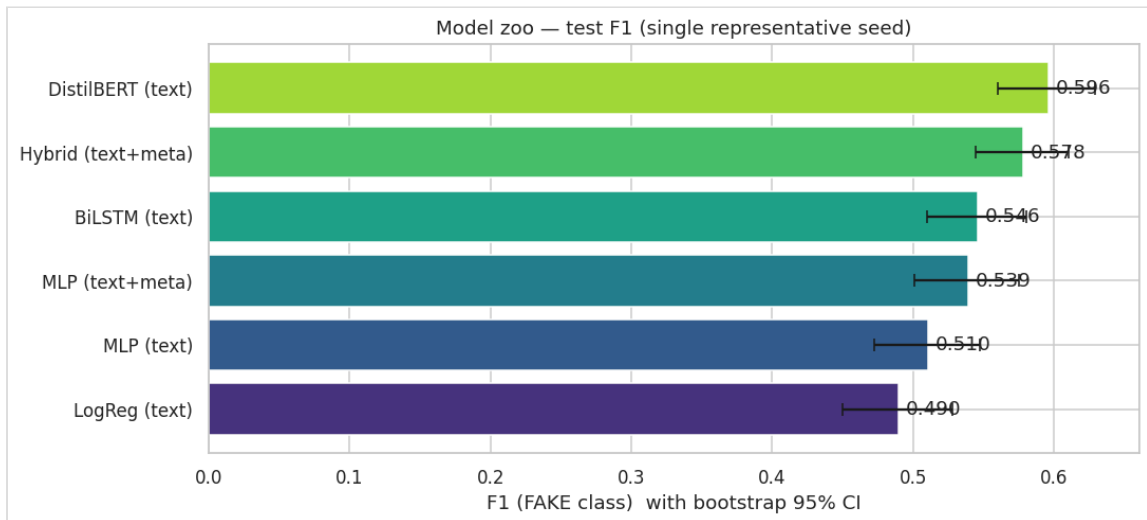


Figure 3. Test F1 by model with bootstrap 95% confidence intervals.

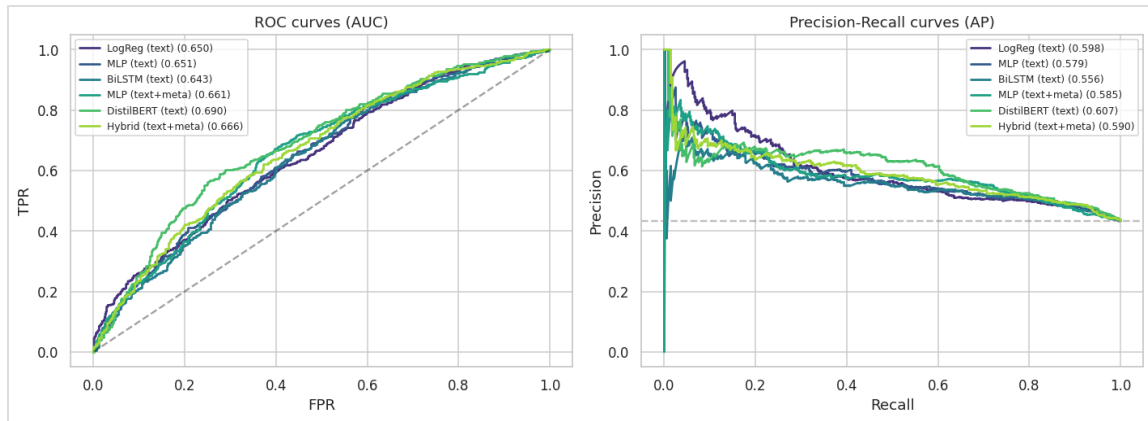


Figure 4. ROC and precision-recall curves for all six models.

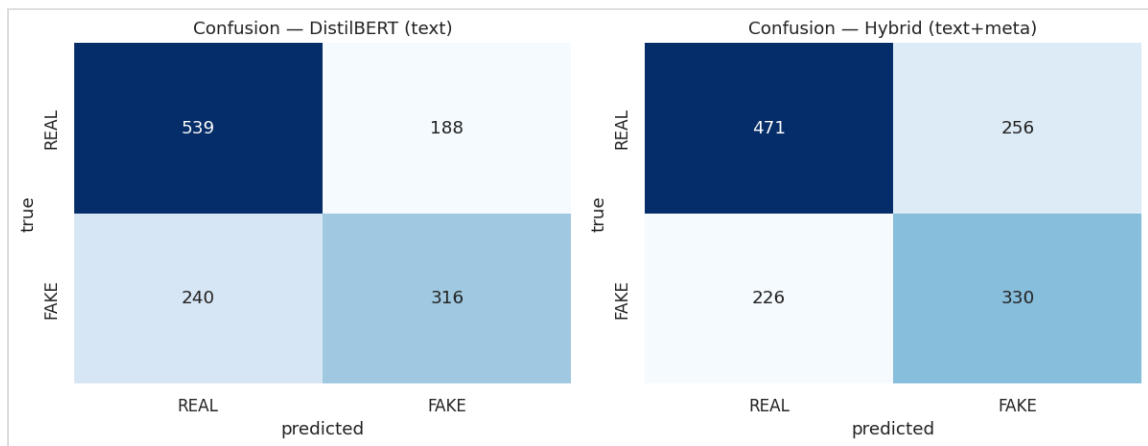


Figure 5. Confusion matrices for the two best models (DistilBERT, Hybrid).

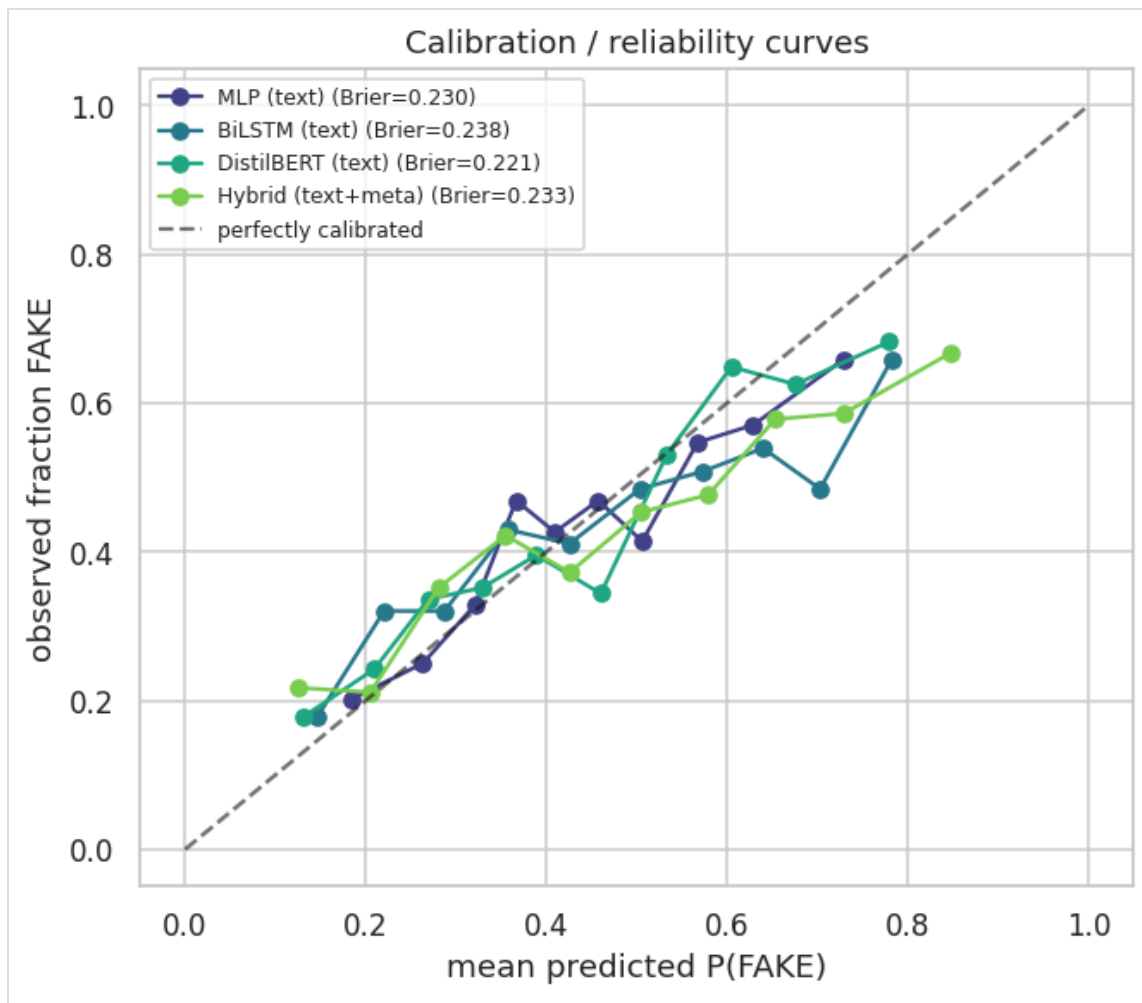


Figure 6. Calibration (reliability) curves; lower Brier score is better.

Analysis

A few things stood out once we put the numbers side by side.

The first is that "there is no signal in the text" was too quick a conclusion. Our from-scratch MLP and LSTM both stall around F1 0.51 to 0.55, which makes it tempting to say the words just don't carry the answer. But DistilBERT, reading the exact same statements, reaches 0.60. The difference is not the input, it is what the model already knows. A network pretrained on a large corpus brings outside knowledge that ten thousand short examples cannot teach from scratch. Experiment 3 supports this from the other direction: swapping in pretrained word vectors (GloVe) did nothing, but a pretrained model helped. The useful knowledge lives in the whole encoder, not just the embeddings.

The second is that the speaker matters more than the sentence. Metadata on its own beats every text model on AUC. We are upfront about the catch: LIAR's credit-history counts include the current statement, so they partly leak how truthful this speaker usually is. That makes the metadata result as much a measure of reputation as of content, which is why we treat the clean text-only DistilBERT number as the honest headline. The practical point still holds, though. Who is speaking is a strong prior, and a real system would use both the text and the track record.

The hybrid lands where you would expect. Adding metadata to the BiLSTM lifts it from 0.55 to 0.58, the best of the from-scratch models, but it cannot catch DistilBERT because its text half is still a weak encoder. The obvious next step, which we did not have time to run, is to fuse DistilBERT with the metadata instead of the BiLSTM.

The errors are consistent across models. Every model under-detects FAKE, with recall lower than precision, and the mistakes cluster on the borderline labels. The MLP is wrong most often on "false" and "barely-true" statements (error rates around 0.52 to 0.57) and almost never on clear "mostly-true" ones (0.19). About a quarter of the test set is missed by both the MLP and the LSTM, which is a fair estimate of the floor: these are claims you cannot settle from eighteen

words without actually checking the facts. The confident mistakes make this concrete. The false negatives we looked at were things like a specific figure for oil reserves or household income, statements that read as completely neutral and can only be caught with outside evidence.

Finally, on whether these numbers are any good: F1 around 0.6 looks modest at first glance, but for LIAR it is where the published work sits. Reported binary results on this dataset cluster in the low-to-mid 60s for accuracy, and numbers much higher than that usually mean the model is using leaked or external information. We would rather report an honest 0.67 with confidence intervals than an inflated figure we could not defend.

Reproducibility

The notebook runs top to bottom on a single GPU and downloads the data itself. It uses the official splits, fits all preprocessing on the training set only, fixes all random seeds, and backs the comparisons with multi-seed averages, McNemar's test and bootstrap confidence intervals.

Contributions

To be finalised before submission — per-member contributions will be listed here.